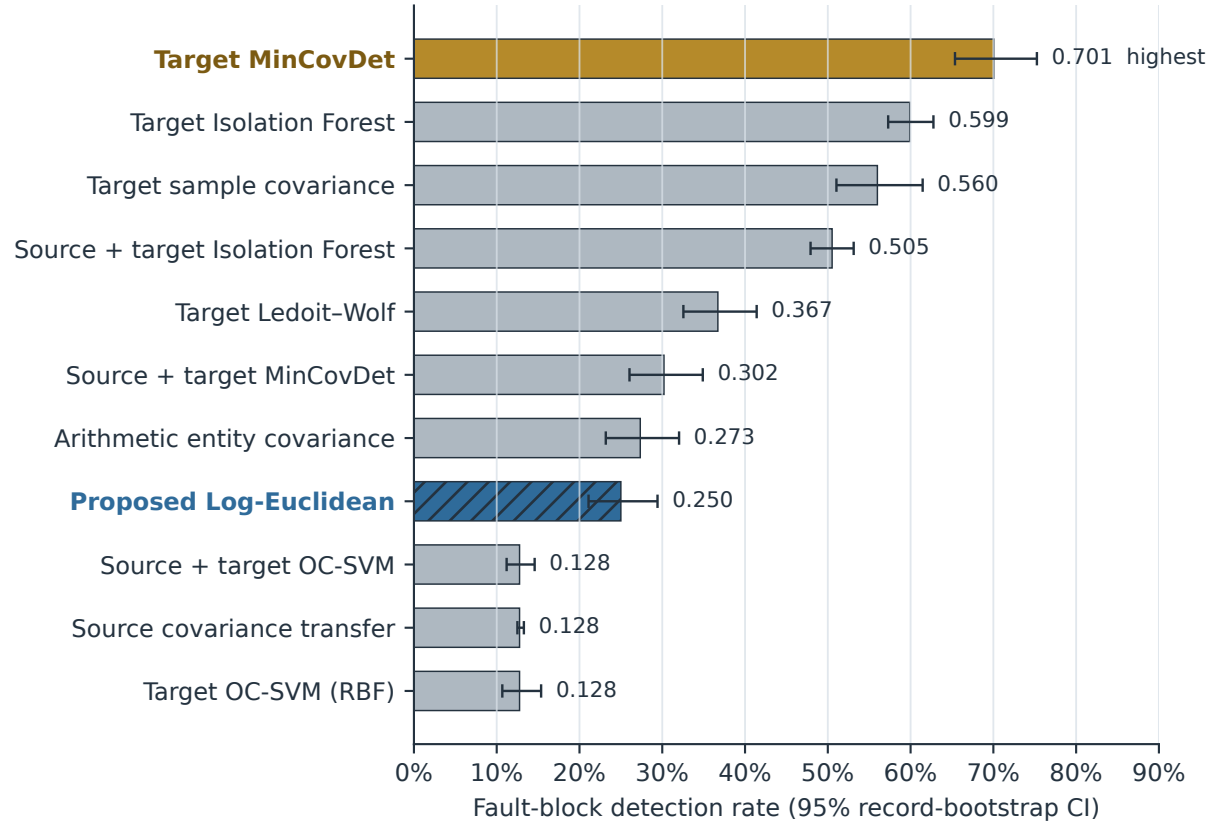


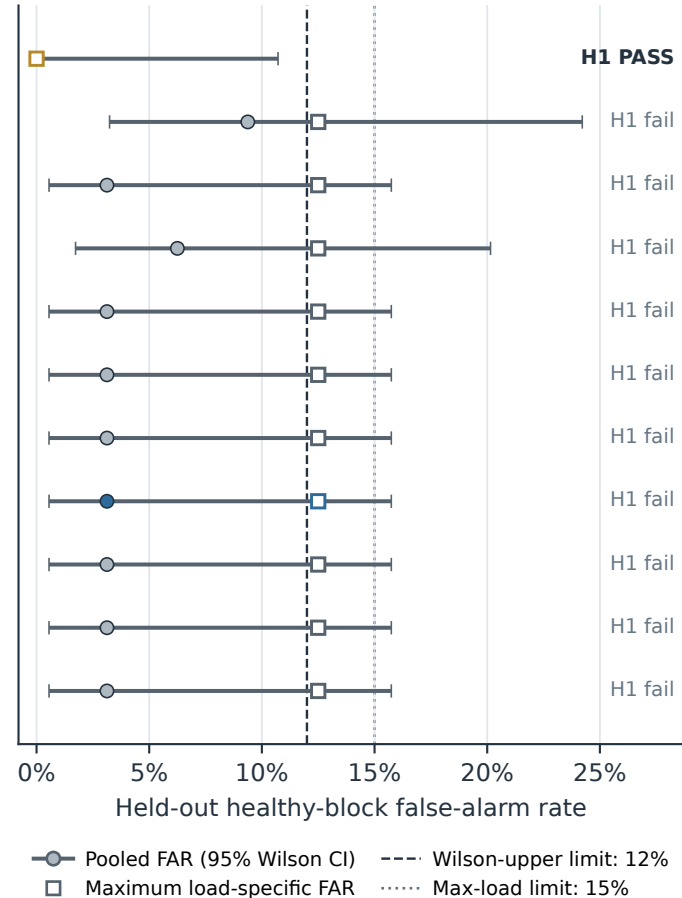
External validation performance by method

48 fault records (384 blocks); four held-out healthy loads (32 blocks). Intervals are descriptive at the recorded-data level.

(a) Fault detection



(b) Healthy false alarms and H1



H1 passes only when the pooled Wilson upper bound $\leq 12\%$ and the maximum held-out-load FAR $\leq 15\%$. Gold marks the observed detection leader; hatched blue marks the proposed method.